

MATH 2030: ODE

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Abstract

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Chapter 1

Linear and First Order ODE

Lecture 1

1.1 Linear ODE of Order 1

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Remark.

$$\frac{dy}{dt} + p(t)y = g(t)$$

Where y is the solution and the rest are going to be the coefficients and $g(t)$ is the forcing term. If $g(t) = 0$ then this is homogeneous

Example.

$$\begin{aligned}y'(t) + 2y(t) &= te^{-2t} \\ e^{2t}y'(t) + 2y(t)e^{2t} &= t\end{aligned}$$

We see that this is equivalent to

$$\frac{d}{dt}(e^{2t}y(t)) = t$$

Integrating we get that

$$\begin{aligned}e^{2t}y(t) &= \frac{1}{2}t^2 + C \\ y(t) &= e^{-2t}\frac{t^2}{2} + Ce^{-2t}\end{aligned}$$

This is going to be the general solution of the original equation.

Remark. In general, we would like to solve the following

$$\frac{d}{dt}y(t) + p(t)y(t) = g(t)$$

We will multiply by $\mu(t)$ which is the integrating factor. We thus have

$$\mu(t)\frac{dy}{dt} + \mu(t)p(t)y(t) = \mu(t)g(t)$$

We wish to write this as a factor of $y(t)$ and this is going to give $\mu(t)g(t)$ We thus have that

$$\begin{aligned}\frac{d}{dt}(\mu(t)y(t)) &= \mu(t)g(t) \\ \mu(t)y'(t) + \mu'(t)y(t) &= \mu(t)g(t)\end{aligned}$$

We choose $\mu(t)$ s.t. $\mu'(t) = p(t)\mu(t)$. Thus we have two equations and we will have that $\mu(t) = e^{\int p(t)dt}$. Integrating we get that

$$\int (\mu(t)y(t))' dt = \int \mu(t)g(t)dt + C$$

$$y(t) = \frac{1}{\mu(t)} \int \mu(t)g(t)dt + C, \mu(t) = e^{\int p(t)dt}$$

Example.

$$t^3 y' + 4t^2 y = t^{-3} e^t$$

$$\mu(t) = e^{\int 4t^{-1} dt}, y' + 4t^{-1} y = t^{-3} e^t$$

$$\mu(t) = t^4$$

We thus have that

$$y(t) = \frac{1}{t^4} \int t e^t dt + \frac{C}{t^4}$$

Example. Find the solution s.t. $y(1) = 0$

$$y(t) = \frac{1}{t^4} (t-1)e^t + \frac{c}{t^4}$$

$$y(1) = \frac{1}{1} (1-1)e^1 + \frac{c}{1} = 0$$

Thus $c = 0, y(t) = \frac{1}{t^4} (t-1)e^t$

Example.

$$y' - \frac{3}{2}y = 3t + 2e^t, y(0) = y_0$$

$$\mu(t) = e^{\int p(t)dt} = e^{\int -\frac{3}{2}dt} = e^{-\frac{3}{2}t}$$

From here we get

$$\frac{d}{dt}(e^{-\frac{3}{2}t}y(t)) = e^{-\frac{3}{2}t}(3t + 2e^{2t})$$

$$e^{-\frac{3}{2}t}y(t) = \int e^{-\frac{3}{2}t}(3t + 2e^{2t})dt$$

$$y(t) = -\frac{4}{3} - 4e^t - 2t + ce^{\frac{3}{2}t}$$

WE have that

$$y(0) = -\frac{4}{3} - 4 + C = -\frac{16}{3} + C = y_0 \Rightarrow C = y_0 + \frac{16}{3}$$

$$y(t) = -\frac{4}{3} - 4e^t - 2t + (y_0 + \frac{16}{3})e^{\frac{3}{2}t}$$

How does $y(t)$ behave as it goes $\pm \infty$.

$$\lim_{t \rightarrow \infty} = \text{inf and } -\text{inf} \Leftrightarrow y_0 + \frac{16}{3} \leq 0$$

$$\lim_{n \rightarrow -\infty} y(t) = \text{inf}$$

Lecture 2

Last time we discussed the solution to the initial value problem

$$y' + p(t)y = g(t), y(0) = y_0$$

Because it is of order one and linear, it is quite simple to find a general solution.

1.2 Uniqueness of Solution

Theorem 1.2.1. Assume that our coefficient $p(t)$ and our forcing term $g(t)$ are continuous functions on some interval $I = (a, b)$ s.t. $t_0 \in (a, b)$. Then the problem above has a unique solution on the interval I

Proof. For existence, let $y(t) = \frac{1}{\mu(t)} \int \mu(t)g(t) + C$ where $\mu(t) = \int_{t_0}^t p(t)dt$

Last time:

$$y' + p(t)y = g$$

$$y(t_0) = \frac{C_0}{e^{\mu(t_0)}} = C_0$$

Therefore we have that $C_0 = y_0$. Given that $p(s), g(s)$ are both differentiable based on the definitions, all of it is differentiable.

For uniqueness, if $y_1(t), y_2(t)$ are two solutions of the equation, $y_1(t) = y_2(t)$. Thus we wish to show that for the same initial conditions, they have the same equation. Let $h = y_1 - y_2$. We have that

$$y_1' + p(t)y_1 = g(t)$$

$$y_2' + p(t)y_2 = g(t)$$

$$(y_1 - y_2)' + p(t)(y_1 - y_2) = 0$$

$$h' + p(t)h = 0$$

$$\frac{d}{dt}(e^{\int p(t)} h(t)) = 0$$

$$e^{\int p(t)} [h'(t) + p(t)h(t)] = 0$$

This implies that we have that

$$\int_{t_0}^t \frac{d}{dt}(e^{\int p(t)} h(t)) = 0$$

$$\mu(t)h(t) - \mu(t_0)h(t_0) = 0$$

But $h(t_0) = 0 \Rightarrow \mu(t)h(t) = 0 \Rightarrow h(t) = 0$. Therefore we see that they have to be the same points.

Example. Nonlinear Example

$$y' - \frac{y^1}{3} = 0, y(0) = 1 > 0$$

$$y' = y^{\frac{1}{3}} \equiv y'/y^{\frac{1}{3}} = 1$$

$$\int y'(t)/y^{\frac{1}{3}} dt = \int dt + C$$

$$\frac{3}{2}y^{\frac{2}{3}}(t) = t + C$$

$$C = \frac{3}{2}, y(t) = \left(\frac{2}{3}t + 1\right)^{\frac{3}{2}}$$

Let us change it so that our initial condition is $y(0) = 0$. We will get two solutions with one of them being

$$y(0) = 0$$

This gives a non-uniqueness condition with continuous coefficients.

Remark. Separable Equations:

Example.

$$\frac{dy}{dx} = \frac{3x^2 - 1}{3 + 2y}, y = y(x)$$

This is nonlinear in terms of y due to the $\frac{1}{y}$ term present.

$$\int (3 + 2y)dy = \int (3x^2 - 1)dx$$

$$3y + y^2 = x^3 - x + C$$

This is written as an implicit solution which we can also move everything on the left hand side and equate it to 0. Solving this we get

$$y = \frac{-3 \pm \sqrt{9 - 4(x - x^3) - C}}{2} = \frac{-3 \pm \sqrt{C + 4x^3 - 4x}}{2}$$

where $\overline{C} \in \mathbb{R}$ and $\overline{C} + 4x^3 - 4x > 0$. We can then specify $\overline{C}$ s.t. $y(0) = 0$. We must have

$$y(0) = -\frac{3}{2} \pm \sqrt{\overline{C}}/2$$

Our solution is determined by the value of y_0 . Thus $\sqrt{\overline{C}}$ changes with respect to the value of y_0

For an initial value of $y_0 = -2$, we will have that $\overline{C} = 1$. To find the interval in which the solution is valid, it is determined by the square root. Thus we can find the solution for the interval where

$$4x^3 - 4x + 1 > 0$$

Example.

$$t^2 y' + 2 + y - y^3 = 0, t > 0$$

The problem with this is not separable, however, we can transform it into a linear equation. We can set $v = y^{1-3} = y^{-2} \equiv y = v^{-\frac{1}{2}}$

$$y' = -\frac{1}{2}v^{-\frac{3}{2}}v'(t)$$

$$-\frac{1}{2}t^2v^{-\frac{3}{2}}v'(t) + 2tv^{-\frac{1}{2}} - v^{-\frac{3}{2}} = 0$$

Multiple by $\frac{v^3}{2}$

$$-\frac{1}{2}t^2v'(t) + 2tv - 1 = 0$$

This has become linear in terms of v and we can solve it using methods of finding $\mu(t)$

$$v'(t) - \frac{4v}{t} = -\frac{2}{t^2}$$

We get that

Lecture 3

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Remark. Recall the equation

$$\begin{aligned}\frac{dy}{dt} &= ry \ln\left(\frac{k}{y}\right), r, k > 0 \\ u = \ln\left(\frac{k}{y}\right) &\Rightarrow \frac{k}{y} = e^u \Rightarrow y = ke^{-u} \\ \frac{dy}{dt} &= -ke^{-u} \frac{du}{dt} \\ -ke^{-u} \frac{du}{dt} &= rke^{-u}u \\ \frac{du}{dt} &= -ru \\ u(t) &= e^{-rt}, \text{ general solution}\end{aligned}$$

We can then substitute back to get

$$y(t) = ke^{-C \exp(rt)}$$

We have $y(0) = y_0 > 0$,

$$y(0) = ke^{-C} = y_0$$

$$C = \ln\left(\frac{k}{y_0}\right)$$

Thus we have that

$$y(t) = k \exp\left(-\ln\left(\frac{k}{y_0}\right) \exp(rt)\right)$$

1.3 Nonlinear First Order

Remark. Exact Equations with special form

Example.

$$2xy^2 + 2y + (2x^2y + 2x)y' = 0, y = y(x) \quad (1.1)$$

This is very nonlinear and can be solved by making a substitution as

$$\frac{d}{dx}\varphi(x, y) = 0$$

for some function $\varphi(x, y)$. In this case, the equation present is called exact. We can apply the chain rule to give that

$$\frac{d}{dx}\varphi(x, y) = \varphi_x + \varphi_y \frac{dy}{dx} = 0$$

We will have that

$$\varphi_x = 2xy^2 + 2y, \varphi_y = 2x^2y + 2x + h'(y)$$

We can integrate and get that

$$\varphi(x, y) = \int 2xy^2 + 2y dx = x^2y^2 + 2xy + h(y)$$

$$\varphi(x, y) = \int \varphi_y dy = x^2y^2 + 2xy + h(y)$$

Hence we see that the term holds when $h = 0$

$$\frac{d}{dx}(y^2x^2 + 2xy) = 0$$

$$x^2y^2 + 2xy = C, C \in \mathcal{R}$$

This defines the function of y implicitly.

Remark. General setting:

$$P(x, y) + Q(x, y) \frac{dy}{dx} = 0$$

We express this as

$$\frac{d}{dx}\varphi(x, y) = 0$$

where

$$\varphi_x + \varphi_y \frac{dy}{dx} = 0$$

Thus we observe that we must have the case that

$$\varphi_x = P, \varphi_y = Q$$

$$\Rightarrow \varphi_{xy} = P_y, \varphi_{yx} = Q_x \Rightarrow \varphi_{xy} = \varphi_{yx}$$

This implies then that

$$P_y = Q_x$$

This is a guaranteed solution. If it is not exact, we must check a different case for the solution.

Definition 1.3.1.

$$P(x, y) + Q(x, y) \frac{dy}{dx} = 0 \quad (1.2)$$

is exact iff $P_y = Q_x \equiv \frac{d}{dx}\varphi(x, y) = 0$ which is the same as our initial equation given in (1.2).

Example.

$$(y \exp(2xy) + x) dx + bx \exp(2xy) dy = 0$$

We can find b s.t. the equation is exact. We will have that

$$P_y = 2xye^{2xy} + ye^{2xy}$$

$$Q_x = be^{2xy} + 2bxye^{2xy}$$

$b = 1$ Therefore we have to find the condition that

$$\varphi_x = P, \varphi_y = Q$$

$$\varphi_y = xe^{2xy} \Rightarrow \varphi(x, y) = \int xe^{2xy} dy + h(x)$$

$$\varphi(x, y) = \frac{1}{2}e^{2xy} + h(x) = ye^{2xy} + x \Rightarrow h'(x) = x$$

$$h(x) = \frac{1}{2}x^2$$

Our equation becomes

$$\frac{d}{dx} \left(\frac{1}{2}e^{2xy} + \frac{x^2}{2} \right) = 0$$

Therefore our solution becomes

$$\frac{1}{2} (e^{2xy} + x^2) = C$$

$$e^{2xy} + x^2 = C$$

Remark. Assume that

$$P(x, y) + Q(x, y) \frac{dy}{dx} = 0$$

is not exact where $P_y \neq Q_x$. In this case, we can try to multiply by $r(x, y)$ such that the equation does become exact where

$$r(x, y)P(x, y) + r(x, y)Q(x, y) \frac{dy}{dx} = 0$$

We can choose $r(x, y)$ where the equation becomes exact. This means that

$$(rP)_y = (rQ)_x$$

We must have that

$$r_y P + r P_y = r_x Q + r Q_x$$

We need to find $r(x, y)$ such that the above condition holds. Next time we will check a theorem for this condition.

Lecture 4

1.4 Existence and Uniqueness of Solutions

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For existence and uniqueness

$$y' = f(t, y), y(0) = y_0$$

WLOG, $y(0) = 0$

Remark. If $y(t)$ solves the equation $y' = f(t, y)$, then $\tilde{y}(t) = y(t + t_0) + y_0$ solves

$$\tilde{y}'(t) = f(t + t_0, \tilde{y}(t) + y_0) = \tilde{f}(t, \tilde{y}(t))$$

where

$$\tilde{y}(0) = 0$$

We now ask under what conditions is guaranteed to have a solution. This is useful for simulations to guarantee a unique solution.

Theorem 1.4.1. If f, f_y are continuous on the rectangular denoted

$$R = \{|t| \leq a, |y| \leq b\}$$

Thus we can draw the following

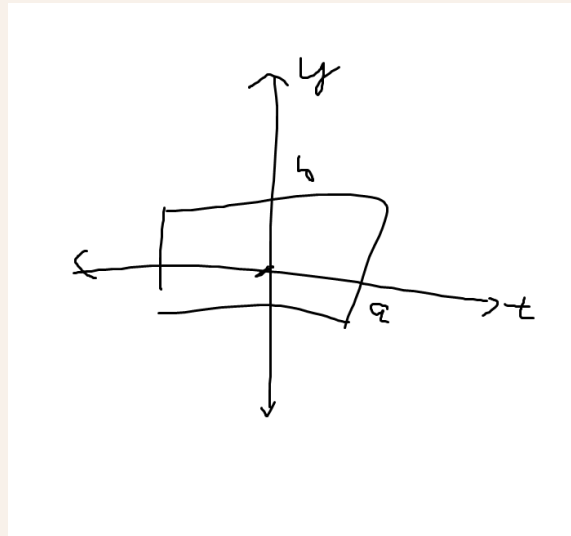


Figure 1.1

Then the conclusion is that there exists $h > 0$ where the problem has a solution where the solution doesn't have to be defined on the whole rectangular but rather restrict the domain to smaller than h . Thus, we have a unique solution of $(y(t))$

$$y' = f(t, y)$$

$$y(0) = 0$$

for $y(t)$ defined on $|t| < h$

Proof. Assume that $\phi(t)$ solves $y' = f(t, y), y(0) = 0$. Therefore we know that

$$\phi'(t) = f(t, \phi(t))$$

We can integrate from 0 to any time t

$$\phi(t) - \phi(0) = \int_0^t f(s, \phi(s)) \, ds$$

we can integrate on $[0, t]$ using $\phi(0) = 0$. Therefore we have that

$$\phi(t) = \int_0^t f(s, \phi(s)) \, ds$$

The method to solve this is through successive approximations (Picard):

Let $\phi_0(t) = 0$, we can now define $\phi_1(t)$ to be

$$= \int_0^t f(s, \phi_0(s)) \, ds$$

we continue this process to get $\phi_2(t)$ to be

$$\int_0^t f(s, \phi_1(s)) \, ds$$

Thus we get the form

$$\phi_{n+1}(t) = \int_0^t f(s, \phi_n(s)) \, ds$$

We can write this notation as (P_n) . Now $P(n)$ defines a sequence of functions ϕ_n . We claim that

$$\lim_{n \rightarrow \infty} \phi_n \rightarrow \phi, [-h, h]$$

for some h uniformly in t . Once we know $P(n)$, we can take the limits of both sides. We must have that

$$\phi_{n+1} \rightarrow \phi$$

and

$$\int_0^t f(t, \phi_n(s)) \, ds \rightarrow \int_0^t f(s, \phi(s)) \, ds$$

Therefore it holds that

$$\begin{aligned} \phi(t) &= \int_0^t f(t, \phi(s)) \, ds \Rightarrow \phi'(t) = f(t, \phi(t)) \\ \phi(0) &= 0 \end{aligned}$$

Remark. $f_y(t, y)$ is continuous on R where R is a compact domain. Based on a well known theorem (which one?), then f_y is bounded on our interval R , $R = |t| \leq a, |y| \leq b$. In other words

$$|f_y(t, y)| \leq M \text{ on } R$$

From here we want to estimate the difference

$$|f(t, y_1) - f(t, y_2)|$$

We can use the Minimum Value Theorem and the above:

$$|f(t, y_1) - f(t, y_2)| = |f_y(t, y^*)| \cdot |y_1 - y_2| \leq M|y_1 - y_2| \forall (t, y_1), (t, y_2) \in R$$

We now have that

$$|f(t, y_1) - f(t, y_2)| \leq M|y_1 - y_2|$$

We can now show the sequence ϕ_{n+1} converges.

$$\phi_{n+1} = \phi_n + (\phi_2 - \phi_1) + (\phi_3 - \phi_2) + \cdots + (\phi_{n+1} - \phi_n)$$

$$\phi_{n+1} = \phi_1 + \sum_{k=1}^n \phi_{k+1}(t) - \phi_k(t)$$

We have now shown that ϕ_{n+1} converges to a value. Thus ϕ_n is equivalent to $\sum_{k=1}^n (\phi_{k+1} - \phi_k)$ converging. Thus, the limit is obvious as shown above. Now notice that the ϕ_n are defined inductively. With the series, we can estimate what each term is in the series and can tell convergence.

We can write a second claim:

$$|\phi_{n+1}(t) - \phi_n(t)| \leq \frac{M^{n+1}}{(n+1)!} |t|^{n+1}, \forall n, |t| \leq h$$

Proof. The proof of this is by induction on k . Assume that it holds for $k - 1$, Then it holds for all k

$$|\phi_{k+1}(t) - \phi_k(t)| = \left| \int_0^t f(s, \phi_k(s)) - f(s, \phi_{k-1}(s)) \, ds \right|$$

where $t > 0$. Then we have that the above holds that

$$\begin{aligned} &\leq \int_0^t |f(s, \phi_k(s)) - f(s, \phi_{k-1}(s))| \, ds \\ &\leq M \int_0^t |\phi_k(s) - \phi_{k-1}(s)| \, ds \end{aligned}$$

This is good because we can use our inductive hypothesis and apply this to k :

$$\leq \frac{M^{k+1}}{k!} \int_0^t S^k \, ds = \frac{M^{k+1}}{k!} \cdot \frac{t^{k+1}}{k+1}$$

Thus by induction this formula holds. Now for the base case, we can apply this for $k = 1$ and $k = 0$ by estimating $\phi_s(s)$ and using the same method as we did above. (Left as an Exercise!) ■

From here, we would like to show that the series converges. We can use Claim 2 to show our main claim

$$\begin{aligned} \phi_{n+1} &= \phi_n + \sum_{k=1}^n \phi_{k+1}(t) - \phi_k(t) \\ \sum_{k=1}^n |\phi_{k+1}(t) - \phi_k(t)| &\leq \sum_{k=1}^n \frac{M^{k+1} t^{k+1}}{(k+1)!} \end{aligned}$$

Thus the series converges if we use the ratio test.

$$\lim_{k \rightarrow \infty} \frac{a_k}{a_{k-1}} < 1, a_k = \frac{M^{k+1} t^{k+1}}{(k+1)!}$$

Computing this limit we have

$$\lim_{k \rightarrow \infty} \frac{Mt}{k}$$

Recall that t is bounded therefore we have that

$$\frac{Mt}{k} \leq \frac{Ma}{k} = 0$$

Thus, the series converges. ■

Lecture 5

Theorem 1.4.2.

$$IVT \begin{cases} f' = f(t, y), & \text{if } R[|t| \leq a, |y| \leq b]; \\ y(0) = 0 \end{cases}$$

One important thing to note is that

$$f(t, y_1) - f(t, y_2) \leq M|y_1 - y_2| \forall (t, y_1), (t, y_2) \in \mathcal{R}$$

where $|f_y|$ is bounded on $\mathcal{R}$.

Proof. Claim: Uniqueness Let ϕ_1 and ϕ_2 be 2 solutions of IVP. From last lecture we had

$$\phi_1(t) = \int_0^t f(s, \phi_1(s)) \, ds$$

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$$\phi_2(t) = \int_0^t f(s, \phi_2(s)) \, ds$$

Hence it is sufficient to show that $\phi_1 = \phi_2$. Let us define

$$u(t) = |\phi_1(t) - \phi_2(t)|$$

$$U(t) = \int_0^t u(t) \, dt, t > 0$$

Then we must have that

$$U'(t) = u(t) = |\phi_1(t) - \phi_2(t)|$$

On the other hand subtracting $E_1 - E_2$

$$\phi_1(t) - \phi_2(t) = \left| \int_0^t f(s, \phi_1(s)) - f(s, \phi_2(s)) \, ds \right|$$

Remember that for $t > 0$,

$$\left| \int_a^b h(s) \, ds \right| \leq \int_a^b |h(s)| \, ds$$

for $a < b$ Therefore

$$|\phi_1(t) - \phi_2(t)| \leq \int_0^t |f(s, \phi_1(s)) - f(s, \phi_2(s))| \, ds \leq M \int_0^t |\phi_1(s) - \phi_2(s)| \, ds$$

Hence we have that

$$u(t) \leq M \int_0^t u(s) \, ds, t > 0, u \geq 0$$

However we have that

$$\frac{M \int_0^t u(s) \, ds}{U(t)} \leq MU(t)$$

Therefore

$$\begin{cases} U'(t) \leq MU(t); \\ U(0) = 0 \end{cases}$$

$$U'(t) - MU(t) \leq 0 \Rightarrow (e^{-Mt}U(t))' \leq 0$$

$$e^{-Mt}(U(t)) - U(0) \leq 0$$

$$\Rightarrow e^{-Mt}U(t) \leq 0$$

but

$$U \geq 0 \Rightarrow U(t) = 0 \Rightarrow \phi_1 = \phi_2$$

■

Chapter 2

Linear Equations of Order 2

$$IVT \begin{cases} y'' + p(t)y' + q(t)y = g(t); \\ y(t_0) = y_0, y'(t_0) = y'_0 \end{cases}$$

Why do we need two parameters?

$$y'' = 0 \Rightarrow y = at + b$$

which are 2 parameters.

Theorem 2.0.1. Assume that $p(t), q(t), g(t)$ are continuous on $I = (a, b)$ s.t. $t_0 \in I$ then the IVP has a unique solution

2.0.1 Constant Coefficient

$$ay'' + by' + cy = 0$$

The idea of a solution is to guess the general form of the solution by looking down a degree:

$$by' + cy = 0$$

$$y(t) = Me^{-\frac{c}{b}t}$$

$$y(t) = e^{rt}, r \in \mathcal{R}$$

where r is determined by coefficients. Let $y(t) = e^{rt}$, then

$$y'(t) = re^{rt}, y'' = r^2e^{rt}$$

Plugging in we have

$$ar^2e^{rt} + bre^{rt} + ce^{rt} = 0$$

$$e^{rt}(ar^2 + br + c) = 0$$

$$\Rightarrow ar^2 + br + c = 0$$

which is defined to be our characteristic equation

Case 1 The C-E has 2 distinct real roots $r_1, r_2 \in \mathcal{R}, r_1 \neq r_2$

$$y_1(t) = e^{r_1t}, y_2(t) = e^{r_2t}$$

Example.

Case 2 The C-E has 2 complex roots $r_{1,2} = \lambda \pm \mu i$

$$y_1(t) = e^{(\lambda + \mu i)t} = e^{\lambda t} e^{\mu i t} = e^{\lambda t} (\cos \mu t + i \sin \mu t)$$

$$y_2(t) = e^{(\lambda - \mu i)t} = e^{\lambda t} (\cos \mu t - i \sin \mu t)$$

Now we have

$$\frac{y_1 + y_2}{2} = e^{\lambda t} \cos \mu t = Y_1(t)$$

$$\frac{y_1 - y_2}{2i} = e^{\lambda t} \sin \mu t = Y_2(t)$$

We conclude that we have two solutions

$$Y_1(t) = e^{\lambda t} \cos \mu t$$

$$Y_2(t) = e^{\lambda t} \sin \mu t$$

Lecture 6

Lecture 7

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2.1 Wronskian

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$$y'' + p(t)y' + q(t)y = 0$$

on interval $I = (a, b)$

Theorem 2.1.1. Let y_1, y_2 be two solutions of the above equation. Let

$$W(t) = \begin{pmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{pmatrix} = (y_1 y_2' - y_1' y_2)(t)$$

This determinant is determined by y_1, y_2 where

$$W[y_1, y_2] = C e^{-\int p(t) dt}$$

for some $C \in \mathbb{R}$ that depends on y_1, y_2 . As a consequence either $C = 0$ or $C \neq 0$ and thus the solutions are dependent or independent. If $y_1 = \lambda y_2 \Rightarrow W[y_1, y_2] = 0$ then they are dependent and vice versa.

Proof. Claim:

$$W'(t) + p(t)W(t) = 0$$

for all $t \in I$. Assume that the claim holds then we have that

$$W(t) = C e^{-\int p(t) dt}$$

for some $C \in \mathbb{R}$. Hence the theorem holds.

$$y_1'' + p(t)y_1' + q(t)y_1 = 0$$

$$y_2'' + p(t)y_2' + q(t)y_2 = 0$$

Let us multiply by y_2 and y_1 respectively and subtract the equations. Thus, we will end up with

$$y_2 y_1'' - y_1 y_2'' + p(t) [y_2 y_1' - y_1 y_2'] = 0$$

We can then show that

$$\begin{aligned} & \frac{d}{dt} [y_2 y_1' - y_1 y_2'] \\ & y_2' y_1' + y_2 y_1'' - y_1' y_2' - y_1 y_2'' \\ & = y_2 y_1'' - y_1 y_2'' \end{aligned}$$

Thus this term is simply just W' which is exactly the form we want. Thus, our end result will get us that

$$W(t) = Ce^{-\int p(t)dt}$$

Note the following are equivalent that:

- y_1, y_2 form a fundamental set of solutions for *.
- y_1, y_2 are linearly independent solutions of *
- $W[y_1, y_2](t_0) \neq 0$ for some $t_0 \in I$
- the third implies this where $W[y_1, y_2](t_0) \neq 0$ for all t

Example. Find the Wronskian of the solutions of:

$$t^2 y'' - t(t+2)y' + (t+2)y = 0$$

$$y'' - \frac{t+2}{t}y' + \frac{t+2}{t^2}y = 0$$

$$W(t) = Ce^{-\int \frac{t+2}{t} dt} = Ce^{\int 1 + \frac{2}{t} dt} = Ce^{t + \ln t^2}$$

$$W(t) = Ce^t t^2$$

Notice that the coefficients are not continuous at the point $t_0 = 0$.

Example. If y_1, y_2 are two solutions of

$$ty'' + 2y' + te^t = 0$$

and $W[y_1, y_2](1) = 2$, find $W[y_1, y_2](5)$. Dividing by t we have

$$W(t) = Ce^{-\int \frac{2}{t} dt} = Ct^{-2}$$

$$C = 2 \Rightarrow W(t) = \frac{2}{t^2}$$

2.2 Characteristic Equation

Consider the equation

$$ay'' + by' + cy = 0$$

Set $y = e^{rt}$, then we have that

$$ar^2 + br + c = 0$$

and let r_1, r_2 be two solutions of the characteristic equation where $r_1 \neq r_2$. Then

$$y_1 = e^{r_1 t}, y_2 = e^{r_2 t}$$

Next we have that $r = \lambda \pm \mu i$. Thus we have the solutions

$$y_1 = e^{\lambda t} \cos(\mu t)$$

$$y_2 = e^{\lambda t} \sin(\mu t)$$

The third case we have is of the solution r as a double root the characteristic equation. Then we have

$$y_1 = e^{rt}$$

$$y_2 = te^{rt}$$

To get such a solution, we can perform a reduction of order: Let $y_1 = e^{rt}$ as a solution. We look for another solution of the form

$$y_2(t) = u(t)y_1(t)$$

The idea is that $u(t)$ will solve the equation for the form

$$u'' + fu' = 0$$

where f is t .

Lecture 8

2.3 Summary of Homogeneous Second Order Equations

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A homogeneous second order equation takes the form:

$$ay'' + by' + cy = 0$$

Setting $y = e^{rt}$, we obtain the characteristic equation:

$$ar^2 + br + c = 0$$

Cases:

- $r_1, r_2 \in \mathbb{R}$ and $r_1 \neq r_2$: The solutions are $y_1 = e^{r_1 t}$, $y_2 = e^{r_2 t}$.
- $r = \lambda \pm \mu i$ (complex roots): The solutions are $y_1 = e^{\lambda t} \cos(\mu t)$ and $y_2 = e^{\lambda t} \sin(\mu t)$.
- r is a double root: The solutions are $y_1 = e^{rt}$ and $y_2 = te^{rt}$.

2.4 Reduction of Order

Given a homogeneous second order equation:

$$y'' + p(t)y' + q(t)y = 0$$

and one known solution y_1 , we look for another solution of the form:

$$y_2(t) = u(t)y_1(t)$$

Differentiating, we obtain:

$$\begin{aligned} y_2' &= u'y_1 + uy_1' \\ y_2'' &= u''y_1 + 2u'y_1' + uy_1'' \end{aligned}$$

Substituting into the equation and simplifying, we get:

$$u''y_1 + 2u'y_1' + uy_1'' + p(t)(u'y_1 + uy_1') + q(t)uy_1 = 0$$

which simplifies to:

$$u''y_1 + (2y_1' + py_1)u' = 0$$

This equation can be solved for $u(t)$, leading to y_2 .

2.5 Examples

Example. Given a homogeneous second order equation:

$$ay'' + by' + cy = 0$$

and a known solution $y_1 = e^{rt}$, the characteristic equation is:

$$ar^2 + br + c = 0$$

If r is a double root, then:

$$r = \frac{-b}{2a}$$

Differentiating y_1 , we obtain:

$$y_1' = re^{rt}, \quad y_1'' = r^2e^{rt}$$

Substituting into the differential equation:

$$2re^{rt} + \frac{b}{a}e^{rt} = e^{rt} \left(2r + \frac{b}{a} \right) = 0$$

which holds by definition of r .

Next, we assume $y_2 = u(t)e^{rt}$ and differentiate:

$$y_2' = u'e^{rt} + ure^{rt}$$

$$y_2'' = u''e^{rt} + 2u're^{rt} + ur^2e^{rt}$$

Substituting into the original equation:

$$e^{rt}u'' = 0 \Rightarrow u'' = 0$$

Integrating twice:

$$u' = C_1, \quad u = C_1t + C_2$$

Choosing $C_1 = 1$ and $C_2 = 0$, we obtain $u = t$, so:

$$y_2 = te^{rt}$$

Thus, the general solution is:

$$y = C_1e^{rt} + C_2te^{rt}$$

Chapter 3

Non-Homogeneous

3.1 Non-Homogeneous Equations

Theorem 3.1.1. Let p, q , and g be continuous on I . Every solution $y(t)$ of $*$ is of the form:

$$y(t) = C_1 y_1 + C_2 y_2 + y_p(t)$$

where y_1, y_2 is a fundamental set of solutions of:

$$y'' + p(t)y' + q(t)y = 0 \quad (*)_h$$

Then $y_p(t)$ is one solution of $*$, and $C_1, C_2 \in \mathbb{R}$.

Let $\tilde{y}(t) = C_1 y_1 + C_2 y_2$. Then:

$$\tilde{y}'' + p(t)\tilde{y}' + q(t)\tilde{y} = 0 \quad (1)$$

$$y_p'' + p(t)y_p' + q(t)y_p = g(t) \quad (2)$$

Adding (1) and (2), we obtain:

$$y'' + p(t)y' + q(t)y = g(t)$$

Let $y(t)$ be any solution of $*$ other than $y_p(t)$. Then:

$$y'' + p(t)y' + q(t)y = g(t) \quad (1)$$

$$y_p'' + p(t)y_p' + q(t)y_p = g(t) \quad (2)$$

Subtracting (2) from (1), we get:

$$\tilde{y} = y - y_p \quad \text{satisfies (1) - (2)}$$

which simplifies to:

$$\tilde{y}'' + p(t)\tilde{y}' + q(t)\tilde{y} = 0$$

Hence:

$$\tilde{y} = C_1 y_1 + C_2 y_2$$

If we need to find solutions for:

$$y'' + p(t)y' + q(t)y = g(t)$$

we must determine:

1. y_1
2. y_2
3. y_p

3.2 Methods for Finding y_p

3.2.1 Method of Undetermined Coefficients

Example. Solve:

$$y'' - 2y' - 3y = 3e^{2t}$$

Assume:

$$y_p(t) = Ae^{2t}$$

Differentiating:

$$y'_p = 2Ae^{2t}, \quad y''_p = 4Ae^{2t}$$

Substituting into the equation:

$$4Ae^{2t} - 4Ae^{2t} - 3Ae^{2t} = 3e^{2t}$$

Solving for A :

$$-3A = 3 \Rightarrow A = -1$$

Thus:

$$y_p(t) = -e^{2t}$$

Example. Solve:

$$y'' + 2y' + 5y = 3 \sin t$$

Assume:

$$y_p(t) = A \sin t + B \cos t$$

Differentiating:

$$y'_p = A \cos t - B \sin t$$

$$y''_p = -A \sin t - B \cos t$$

Substituting into the equation:

$$(-A \sin t - B \cos t) + 2(A \cos t - B \sin t) + 5(A \sin t + B \cos t) = 3 \sin t$$

Rearrange:

$$(4A - 2B - 3) \sin t + (4B + 2A) \cos t = 0$$

Solving:

$$4A - 2B = 3, \quad 4B + 2A = 0$$

$$B = -\frac{3}{10}, \quad A = \frac{3}{5}$$

Thus:

$$y_p = \frac{3}{5} \sin t - \frac{3}{10} \cos t$$

Example. Solve:

$$y'' + y' - 2y = 2t^2$$

Assume:

$$y_p(t) = At^2 + Bt + C$$

Differentiating:

$$y'_p = 2At + B, \quad y''_p = 2A$$

Substituting into the equation:

$$2A + (2At + B) + (-2At^2 - 2Bt - 2C) = 2t^2$$

Grouping terms:

$$(-2A)t^2 + (2A - 2B)t + (2A + B - 2C) = 2t^2$$

Solving:

$$-2A = 2 \Rightarrow A = -1$$

$$2A - 2B = 0 \Rightarrow B = -1$$

$$2A + B - 2C = 0 \Rightarrow C = -\frac{3}{2}$$

Thus:

$$y_p = -t^2 - t - \frac{3}{2}$$

Example. Solve:

$$y'' - 2y' + y = (t + 4)e^t$$

Assume:

$$y_p(t) = (At + B)e^t$$

Note: This assumption does not work because $r = 1$ is a double root of $r^2 - 2r + 1 = 0$. Since te^t, e^t are solutions of the homogeneous equation, we must use reduction of order:

$$y_p(t) = (At^2 + Bt)e^t$$

Lecture 9

3.3 Higher Order ODE

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Example. Using undetermined coefficients we have:

$$y''' + 4y' = t$$

We can first solve the homogeneous equation

$$y''' + 4y' = 0$$

$$r^3 + 4r = 0 \Rightarrow r(r^2 + 4) \Rightarrow r = 0, \pm 2i$$

$$y(t) = 1 + c_1 \cos 2t + c_2 \sin 2t + c_3$$

We now find $y_p = at^2 + bt + c$. Notice that

$$4y' = t \Rightarrow y' = \frac{t}{4} \Rightarrow y = \frac{t^2}{8}$$

$$y''' + 4y' = 0 + 4y' = t$$

Therefore the solution must be of the form

$$y(t) = a_1 t^3 + a_2 t^2 + a_3 t + a_4, \quad y_p(t) = \frac{t^2}{8}$$

The general solution must be

$$c_1 + c_2 \cos 2t + c_3 \sin 2t + \frac{t^2}{8}$$

For the variation of parameters we must have that the particular solution to be of the form

$$y_p(t) = u_1 y_1 + u_2 y_2 + \dots$$

$$u_1' y_1 + u_2' y_2 + y_3' y_3 = 0$$

$$y_1' u_1' + u_2' y_2' + u_3' y_3' = 0$$

$$u_1' y_1'' + u_2' y_2'' + u_3' y_3'' = t$$

...

Example.

$$y''' - y' = t$$

Using variational parameters we have the homogeneous equation

$$r^3 - r = 0 \Rightarrow r(r^2 - 1) \Rightarrow r = 0, \pm 1$$

$$y_h(t) = 1 + c_1 e^t + c_2 e^{-t}$$

We have that

$$u_1' + e^t u_2' + e^{-t} u_3' = 0$$

$$0 + e^t u_2' - e^{-t} u_3' = 0$$

$$0 + e^t u_2' + e^{-t} u_3' = t$$

$$\Rightarrow u_2' = \frac{t}{2} e^{-t}$$

$$\Rightarrow u_3' = \frac{t}{2} e^t$$

$$\Rightarrow u_1' = -t$$

Therefore we have found u_1, u_2, u_3 and now the general equation can be solved for.

$$y_p(t) = -\frac{1}{2}t^2 - \frac{1}{2}te^{-t} - \frac{1}{2}e^{-t} + \frac{1}{2}te^t + \frac{1}{2}e^t$$

$$y_p(t) = -\frac{1}{2}t^2 - \frac{1}{2}(t+1) + \frac{1}{2}(t-1) = -\frac{t^2}{2} - 1$$

Therefore our final solution is of the form

$$y(t) = c_1 + c_2 e^t + c_3 e^{-t} - \frac{t^2}{2} - 1$$

Example.

$$y''' - y = 5e^{2t} + t^2$$

Split this into two equations to solve. Consider the homogeneous

$$y''' - y = 0 \Rightarrow r^3 - 1 = 0 \Rightarrow r = e^{\frac{i(0)+2\pi im}{3}}$$

Thus there are three roots for r s.t

$$y_{p1}(t) = Ae^{2t}, \quad y_{p2}(t) = At^3 + Bt^2 + Ct + D$$

Solving these we have that

$$y_{p1}' = 2Ae^{2t}, y_{p1}''' = 8Ae^{2t}$$

$$\Rightarrow y''' - y = 7Ae^{2t} = 5e^{2t}$$

$$A = \frac{5}{7}$$

For the second one we have

$$y_{p2}' = 3At^2 + 2Bt + C$$

$$y_{p2}''' = 6At + 2B$$

$$y_{p2}''' = 6A$$

$$\Rightarrow 6A - (At^3 + Bt^2 + Ct + D) = t^2$$

$$B = -1$$

$$y(t) = y_h(t) + y_{p1}(t) + y_{p2}(t)$$

Lecture 10

3.4 Euler's Equation

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Euler's equation is of the form

$$x^2 y'' + axy' + by = 0$$

We look for solution of the form $y = x^r, x > 0$. It must satisfy

$$r(r-1) + ar + b = 0$$

For case 1

$$r_1, r_2 \in \mathcal{R}$$

$$y_1 = x^{r_1}, y_2 = x^{r_2}, x > 0 \quad y_1 = |x|^{r_1}, y_2 = |x|^{r_2}$$

For case 2

$$r_1 = r_2 \in \mathcal{R}$$

$$y_1 = |x|^{r_1}, y_2 = |x|^{r_2}$$

For case 3

$$r_1 = \lambda + i\mu \in \mathcal{C} \quad r_2 = \lambda - i\mu$$

$$y_1 = |x|^\lambda \cos(\mu \ln |x|), \quad y_2 = |x|^\lambda \sin(\mu \ln |x|)$$

Consider the series solution on a regular singular point where the singular point here is for $x = 0$.

$$y'' + p(x)y' + q(x)y = 0$$

might be "singular" at some point.

Example. Euler's is written as

$$y'' + \frac{a}{x}y' + \frac{b}{x^2}y = 0 \Rightarrow x = 0 \text{ singular point}$$

x_0 is a regular singular point if:

1. x_0 is a singular point
2. $\lim_{x \rightarrow x_0} (x - x_0)p(x) = a_0$
3. $\lim_{x \rightarrow x_0} (x - x_0)^2 q(x) = b_0$

This implies that

$$p(x) \approx \frac{a_0}{x - x_0}, \quad q(x) \approx \frac{b_0}{(x - x_0)^2}$$

Thus, the above approximates behavior for x close to x_0 where

$$y'' + \frac{a_0}{x - x_0}y' + \frac{b_0}{(x - x_0)^2}y = 0$$

This gives an approximate solution for the problem we have. Hence, we expect the solution of $y'' + p(x)y' + q(x)y = 0$ behaves to the first order for

$$y(x) \approx \tilde{y}(x)$$

s.t. $\tilde{y}$ solves the approximate equation. Hence if the approximate equation has one solution s.t

$$y_1(x) = |x|^r$$

then we expect that for $x > 0$ we expect that $y(x) \approx x^r$

Example.

$$xy'' + y' - y = 0$$

Find one series solution

$$y'' + \frac{1}{y}y' - \frac{y}{x} = 0 \Rightarrow x = 0 \text{ singular}$$

$$\lim_{x \rightarrow 0} \frac{x}{x} = 1 \quad \lim_{x \rightarrow 0} \frac{x^2}{x} = 0$$

therefore x is regular singular. Next, we look for a series solution of the form

$$f(x) = x^r \sum_{n=0}^{\infty} a_n x^n$$

where r is a root of the indicial equation of the approximating Euler's equation

$$x^2 y'' + x y' = 0, \quad a_0 = 1, b_0 = 0$$

$$y = x^r \quad r(r-1) + r = 0$$

$$r^2 - r + r = 0 \Rightarrow r = 0 \Rightarrow 2 \text{ roots}$$

Solutions of the equation will have the form

$$y(x) = x^r = 1 \quad y(x) = \ln x$$

Consider the equation again

$$xy'' + y' - y = 0$$

$$y(x) = x^r \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} (n+4) a_n x^{n+r-1}, \quad y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

Plugging in we have

$$xy'' + y' - y = x \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2} + \sum_{n=0}^{\infty} (n+r) a_n x^{n+r-1} - \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

Lowering the index of the second term and combining like terms

$$\begin{aligned} &= \sum_{n=0}^{\infty} [(n+r)(n+r-1) + (n+r)] a_n x^{n+r-1} - \sum_{n=1}^{\infty} a_{n-1} x^{n+r-1} \\ &= \sum_{n=1}^{\infty} [(n+r)^2 a_n - a_{n-1}] x^{n+r-1} + a_0 r^2 x^{r-1} = 0 \end{aligned}$$

In order for this to be 0 the coefficients must be 0 s.t

$$a_0 r^2 = 0 \Rightarrow a_0 = 0 \text{ or } r = 0$$

$$(n+r)^2 a_n - a_{n-1} = 0 \Rightarrow a_n = \frac{a_{n-1}}{(n+r)^2}$$

We see that $a_0 = 0$ gives a trivial series therefore $r = 0$ Therefore the series is

$$a_n = \frac{a_{n-1}}{n^2}$$

Take $a_0 = 1$ we can generate the series where

$$a_n = \frac{1}{(n!)^2}$$

$$y(x) = \sum_{n=0}^{\infty} \frac{|x|^n}{(n!)^2}$$

Example.

$$2x^2y'' + 3xy' + (2x^2 - 1)y = 0$$

$$y'' + \frac{3x}{2x^2}y' + \frac{2x^2 - 1}{2x^2}y = 0$$

$$\lim_{x \rightarrow 0} xp(x) = \lim_{x \rightarrow 0} \frac{3}{2x}x = \frac{3}{2}$$

$$\lim_{x \rightarrow 0} x^2q(x) = \lim_{x \rightarrow 0} \frac{2x^2 - 1}{2x^2}x^2 = -\frac{1}{2}$$

Approximating we have

$$x^2y'' + \frac{3}{2}xy' - \frac{1}{2}y = 0$$

$$r(r - \frac{1}{2}) + \frac{3}{2}r - \frac{1}{2} = 0 \Rightarrow r_1 = -1, r_2 = \frac{1}{2}$$

$$r_1 - r_2 = -\frac{3}{2}$$

which is not an integer and leads to problems with finding the second solution to the equation. If they don't differ by an integer they have two distinct roots.

$$y(x) = x^r \sum_{n=0}^{\infty} a_n x^{n+r}$$

We have

$$2x^2 \sum_{n=0}^{\infty} (n+r)(n+r-1)x^{n+r-2} + 3x \sum_{n=0}^{\infty} (n+r)x^{n+r-1} + 2x^2 \sum_{n=0}^{\infty} a_n x^{n+r} - \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

Combining terms and adding the x^{n+r} terms we have

$$\sum_{n=0}^{\infty} (2(n+r)(n+r-1) + 3(n+r) - 1) a_n x^{n+r} + \sum_{n=2}^{\infty} 2a_{n-2} x^{n+r} = 0$$

$$\sum_{n=2}^{\infty} [(2(n+r)(n+r) - 1 + 3(n+r) - 1)a_n + 2a_{n-2}] x^{n+r} + \sum_{n=0}^{\infty} (2(n+r)(n+r-1) + 3(n+r) - 1) a_n x^{n+r} = 0$$

We see that these two summations will give us 0 and therefore we can find the recursion relationship.